Unit Test Report

Project	Online Exam Portal (Software Engineering mini project)
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Build	`main` at commit `2f93f79`
Test run	11-11-2026 21:00 IST
Generated by	`docs/reports/generate_reports.py` (do not edit by hand; run the script again)
References	SRS v1.0, SAD v1.0, Software Test Plan v1.0, Test Case Document

1. Summary

Metric	Value
Test functions executed	518
Passed	517
Failed	0
Errors	0
Skipped	1
Subtests (parametrised checks inside tests)	337
Pass rate (of tests run)	100.0%
Total run time	153.6 s
pytest summary line	`517 passed, 1 skipped, 337 subtests passed in 153.60s (0:02:33)`
Result	**PASS**

2. Test environment

Item	Value
OS	Linux 6.18.44-fc-v378

Item	Value
Python	3.13.18
Django / DRF	5.2.18 / 3.18.3
Test runner	pytest 9.1.1 with pytest-django, pytest-cov
Coverage tool	coverage.py 7.17.2 (line + branch, settings in `.coveragerc`)
Database	SQLite (Django test database, created fresh for the run)
Command	`python docs/reports/generate_reports.py` (wraps `pytest --cov --junitxml`)

3. Scope and approach

Unit and integration tests are written with Django's `TestCase` / DRF's `APIClient` and run by pytest. Each Django app owns its tests (`tests.py`, `test_*.py`). Service functions (grading, scoring, shuffling, lockout, audit) are tested directly; API endpoints are tested through HTTP requests against a fresh test database, so serializers, permissions and the database layer are exercised together. Every test class is linked to a Test Plan test case ID (Section 5) and from there to an SRS requirement. The same suite runs on every push and pull request in GitHub Actions (`.github/workflows/django.yml`).

4. Results by module

App	SAD module	Tests	Passed	Failed	Skipped	Time	Result
`accounts`	Auth Service, Admin Service, Audit & Logging	134	134	0	0	51.6 s	Pass
`courses`	Admin Service (courses)	20	20	0	0	14.1 s	Pass
`question_bank`	Question Bank Service	75	75	0	0	9.2 s	Pass
`exam_management`	Exam Scheduling Service	59	59	0	0	3.7 s	Pass
`exam_delivery`	Exam Delivery Service, diagnostics	51	50	0	1	25.4 s	Pass
`evaluation`	Evaluation Service	61	61	0	0	28.7 s	Pass
`reports`	Reporting Service	41	41	0	0	1.7 s	Pass
`web`	Web UI	45	45	0	0	16.5 s	Pass
`config`	Platform (settings, TLS, health check)	32	32	0	0	0.4 s	Pass

5. Results by test class

File	Class	Test case	Requirement	Tests	Passed	Failed	Skipped	Result
`accounts/test_account_management.py`	`AccountManagementTests`	TC-Admin-01	OEP-F-060	26	26	0	0	Pass
`accounts/test_audit_log.py`	`AppendOnlyTests`	TC-Audit-01	OEP-NF-004	5	5	0	0	Pass
`accounts/test_audit_log.py`	`KeyActionLoggingTests`	TC-Audit-01	OEP-NF-004	10	10	0	0	Pass
`accounts/test_audit_log.py`	`RetentionTests`	TC-Admin-03, TC-Audit-01	OEP-F-062, OEP-NF-004	14	14	0	0	Pass
`accounts/test_lockout.py`	`AccountLockoutTests`	TC-Auth-04	OEP-F-004	12	12	0	0	Pass
`accounts/test_lockout.py`	`LockoutAuditTests`	TC-Sec-05	OEP-SR-005	4	4	0	0	Pass
`accounts/test_login.py`	`LoginTests`	TC-Auth-02	OEP-F-002	10	10	0	0	Pass
`accounts/test_login.py`	`TokenSessionTests`	TC-Auth-02	OEP-F-002	4	4	0	0	Pass
`accounts/test_password_hashing.py`	`PasswordHashingTests`	TC-Sec-02	OEP-SR-002	5	5	0	0	Pass
`accounts/test_password_hashing.py`	`SensitiveParameterTests`	TC-Sec-02	OEP-SR-002	1	1	0	0	Pass
`accounts/test_rbac.py`	`EndpointRegistryTests`	TC-Auth-03	OEP-F-003	4	4	0	0	Pass
`accounts/test_rbac.py`	`EveryEndpointRoleCheckTests`	TC-Sec-06	OEP-SR-006	3	3	0	0	Pass
`accounts/test_rbac.py`	`RoleMatrixTests`	TC-Auth-03	OEP-F-003	5	5	0	0	Pass
`accounts/test_session_expiry.py`	`SessionExpiryTests`	TC-Sec-03	OEP-SR-003	12	12	0	0	Pass
`accounts/tests.py`	`UserModelTests`	TC-Auth-01	OEP-F-001	4	4	0	0	Pass
`accounts/tests.py`	`StudentRegistrationTests`	TC-Auth-01	OEP-F-001	13	13	0	0	Pass

File	Class	Test case	Requirement	Tests	Passed	Failed	Skipped	Result
`config/test_health.py`	`HealthEndpointTests`	TC-Ops-01	OEP-NF-003	5	5	0	0	Pass
`config/test_tls.py`	`ProxiedLoginAuditTests`	TC-Sec-01	OEP-SR-001	1	1	0	0	Pass
`courses/tests.py`	`CourseCrudTests`	TC-Admin-02	OEP-F-061	7	7	0	0	Pass
`courses/tests.py`	`CourseMappingTests`	TC-Admin-02	OEP-F-061	12	12	0	0	Pass
`courses/tests.py`	`CourseAccessTests`	TC-Admin-02	OEP-F-061	1	1	0	0	Pass
`evaluation/test_grading_queue.py`	`GradingQueueTests`	TC-Eval-02	OEP-F-041	13	13	0	0	Pass
`evaluation/test_scorecard.py`	`ScorecardTests`	TC-Result-01	OEP-F-050	13	13	0	0	Pass
`evaluation/test_scorecard.py`	`EndToEndTests`	TC-Result-01	OEP-F-050	1	1	0	0	Pass
`evaluation/test_subjective_evaluation.py`	`EvaluateResponseTests`	TC-Eval-03	OEP-F-042	17	17	0	0	Pass
`evaluation/test_subjective_evaluation.py`	`FinalScoreIntegrationTests`	TC-Eval-03	OEP-F-042	4	4	0	0	Pass
`evaluation/tests.py`	`ObjectiveGradingCloseOutTests`	TC-Eval-01	OEP-F-040	1	1	0	0	Pass
`evaluation/tests.py`	`FinalizeAttemptScoreTests`	TC-Eval-04	OEP-F-043	12	12	0	0	Pass
`exam_delivery/test_delivery.py`	`ExamDeliveryTimerTests`	TC-Deliver-01	OEP-F-030	5	5	0	0	Pass
`exam_delivery/test_delivery.py`	`QuestionNavigationAndFlagTests`	TC-Deliver-02	OEP-F-031	4	4	0	0	Pass
`exam_delivery/test_delivery.py`	`AutoSubmitTimeoutTests`	TC-Deliver-03	OEP-F-032	2	2	0	0	Pass
`exam_delivery/test_delivery.py`	`ManualSubmitTests`	TC-Deliver-04	OEP-F-033	2	2	0	0	Pass

File	Class	Test case	Requirement	Tests	Passed	Failed	Skipped	Result
`exam_delivery/test_delivery.py`	`SingleAttemptRuleTests`	TC-Deliver-05	OEP-F-034	2	2	0	0	Pass
`exam_delivery/test_delivery.py`	`PeriodicAutosaveTests`	TC-Deliver-06	OEP-F-035	3	3	0	0	Pass
`exam_delivery/test_delivery.py`	`ReviewRegressionTests`	—	—	14	14	0	0	Pass
`exam_delivery/test_delivery.py`	`ConcurrentStartTests`	TC-Deliver-05	OEP-F-034	5	5	0	0	Pass
`exam_delivery/test_delivery.py`	`TrueFalseOptionsTests`	TC-Deliver-01	OEP-F-030	4	4	0	0	Pass
`exam_delivery/test_diagnostics.py`	`AdminDiagnosticsTests`	TC-Admin-04	OEP-F-063	7	7	0	0	Pass
`exam_management/test_answer_key_protection.py`	`ExamWindowTests`	TC-Sec-04	OEP-SR-004	5	5	0	0	Pass
`exam_management/test_answer_key_protection.py`	`StudentPaperContentTests`	TC-Sec-04	OEP-SR-004	8	8	0	0	Pass
`exam_management/test_answer_key_protection.py`	`StudentApiExposureTests`	TC-Sec-04	OEP-SR-004	2	2	0	0	Pass
`exam_management/tests.py`	`ExamAssemblyTests`	TC-Exam-01	OEP-F-020	14	14	0	0	Pass
`exam_management/tests.py`	`ExamAPITests`	TC-Exam-01, TC-Exam-02, TC-Exam-03, TC-Sec-06	OEP-F-020, OEP-F-021, OEP-F-022, OEP-SR-006	19	19	0	0	Pass
`question_bank/tests.py`	`QuestionModelTests`	TC-QB-01	OEP-F-010	4	4	0	0	Pass
`question_bank/tests.py`	`QuestionAPITests`	TC-QB-01, TC-QB-02, TC-Sec-06	OEP-F-010, OEP-F-011, OEP-SR-006	24	24	0	0	Pass
`question_bank/tests.py`	`QuestionTemplateDownloadTests`	TC-QB-03, TC-Sec-06	OEP-F-012, OEP-SR-006	3	3	0	0	Pass
`question_bank/tests.py`	`QuestionBulkImportTests`	TC-QB-03, TC-Sec-06	OEP-F-012, OEP-SR-006	13	13	0	0	Pass

File	Class	Test case	Requirement	Tests	Passed	Failed	Skipped	Result
`question_bank/tests.py`	`QuestionDetailAPITests`	TC-QB-04, TC-Sec-06	OEP-F-013, OEP-SR-006	16	16	0	0	Pass
`reports/test_breakdown.py`	`QuestionBreakdownTests`	TC-Result-04	OEP-F-053	10	10	0	0	Pass
`reports/test_exam_report.py`	`ExamReportTests`	TC-Result-02	OEP-F-051	13	13	0	0	Pass
`reports/test_export.py`	`ExportTests`	TC-Result-03	OEP-F-052	18	18	0	0	Pass
`web/tests.py`	`StaticFilesTests`	TC-UX-01	OEP-NF-005	1	1	0	0	Pass
`web/tests.py`	`LoginPageTests`	TC-Auth-02	OEP-F-002	10	10	0	0	Pass
`web/tests.py`	`DashboardTests`	TC-UX-01	OEP-NF-005	13	13	0	0	Pass
`web/tests.py`	`TakeExamPageTests`	TC-Deliver-01, TC-Deliver-02	OEP-F-030, OEP-F-031	14	14	0	0	Pass
`web/tests.py`	`SystemHealthPanelTests`	TC-Admin-04	OEP-F-063	7	7	0	0	Pass
`exam_delivery/test_nfr.py`	`NonFunctionalRequirementsTests`	TC-Ops-01, TC-Perf-01, TC-Rel-01	OEP-NF-001, OEP-NF-002, OEP-NF-003	3	2	0	1	Pass
`accounts/test_session_expiry.py`	`IdleTimeoutSettingTests`	TC-Sec-03	OEP-SR-003	2	2	0	0	Pass
`config/test_health.py`	`UptimeSummaryTests`	TC-Ops-01	OEP-NF-003	4	4	0	0	Pass
`config/test_health.py`	`ProbeOnceTests`	TC-Ops-01	OEP-NF-003	4	4	0	0	Pass
`config/test_health.py`	`SqliteLockingTests`	TC-Ops-01	OEP-NF-003	1	1	0	0	Pass
`config/test_tls.py`	`TlsSettingsTests`	TC-Sec-01	OEP-SR-001	9	9	0	0	Pass
`config/test_tls.py`	`ClientIpTests`	TC-Sec-01	OEP-SR-001	5	5	0	0	Pass
`config/test_tls.py`	`TlsRequestTests`	TC-Sec-01	OEP-SR-001	2	2	0	0	Pass
`config/tests.py`	`ProjectSmokeTest`	—	—	1	1	0	0	Pass
`exam_management/test_shuffle_paper.py`	`ShufflePaperTests`	TC-Exam-04	OEP-F-023	11	11	0	0	Pass

File	Class	Test case	Requirement	Tests	Passed	Failed	Skipped	Result
`question_bank/test_grading.py`	`(module-level tests)`	TC-Eval-01	OEP-F-040	15	15	0	0	Pass

`–` marks regression and support tests (review fixes, smoke tests) that are not tied to one test case.

6. Skipped tests

Test	Test case	Reason
`exam_delivery/test_nfr.py::NonFunctionalRequirementsTests::test_tc_perf_01_performance_200_concurrent_requests`	TC-Perf-01	SQLite's shared in-memory test database cannot safely service this 200-writer concurrency test; run it against the production database backend.

7. Failures

None. All executed tests passed.

8. Defects found and fixed during the sprints

Taken from the commit history (commits whose subject starts with `fix`). Each was found by a test or a PR review, fixed, and is now covered by a regression test in the suite above.

Commit	Date	Fix
`96bac9a`	2026-11-10	Fix exam start race and True/False options; mark NF-001 run as pending
`acbb661`	2026-11-04	fix: protect questions with student responses
`937b065`	2026-11-03	fix: address student website review nits and add browser checks (OEP-F-030, OEP-NF-005)
`475c675`	2026-11-03	fix: address TLS review: upload limit, client IP, secret key and debug (OEP-SR-001)
`2017eb4`	2026-11-03	fix: reject a bad idle timeout setting with a clear error (OEP-SR-003)
`9019b82`	2026-11-02	fix: hide exams a student can't open and add STATIC_ROOT (OEP-F-030)
`2510acb`	2026-11-01	fix(exam): isolate option shuffle by question
`37605dd`	2026-11-01	fix: enforce deadline on submitted answers and secure diagnostics (OEP-F-030)

Commit	Date	Fix
`a64ac7d`	2026-10-31	fix: improve admin health diagnostics
`6a33fcd`	2026-10-31	fix: secure exam delivery attempt handling
`2a24683`	2026-10-31	fix: preserve unreachable database diagnostics status
`2718d1d`	2026-10-31	fix: tighten exam delivery authorization and resume handling
`d58aff8`	2026-10-31	fix: address stacked pull request review feedback
`c8bac72`	2026-10-26	fix: restrict question endpoints to staff users
`f83fde8`	2026-10-25	fix: restrict question unlinking to draft exams

9. Slowest tests

Test	Time
`accounts/test_lockout.py::LockoutAuditTests::test_each_lockout_creates_one_entry`	1.53 s
`accounts/test_lockout.py::AccountLockoutTests::test_successful_login_resets_the_count`	1.39 s
`web/tests.py::LoginPageTests::test_correct_password_is_refused_while_locked`	1.05 s
`web/tests.py::LoginPageTests::test_fifth_failure_locks_the_account`	1.00 s
`accounts/test_lockout.py::AccountLockoutTests::test_unknown_account_is_never_reported_as_locked`	0.98 s
`accounts/test_account_management.py::AccountManagementTests::test_admin_accounts_cannot_be_deactivated_here`	0.94 s
`accounts/test_lockout.py::AccountLockoutTests::test_login_works_again_after_the_lock_expires`	0.94 s
`accounts/test_lockout.py::AccountLockoutTests::test_counter_starts_again_after_the_lock_expires`	0.91 s
`accounts/test_lockout.py::LockoutAuditTests::test_lockout_records_account_and_timestamp`	0.85 s
`accounts/test_lockout.py::AccountLockoutTests::test_lock_also_applies_outside_the_login_endpoint`	0.84 s

10. Raw results

- JUnit XML: `raw/junit.xml`
- Coverage: Code Coverage Report, Branch Coverage Report